LOS THOERY ASSIGNMENT -1

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1. What If There Were No Operating System?

Imagine a computer without an operating system. Identify the problems a user would face and investigate how OS features solve them. Demonstrate at least three features using Linux and present a *"Without OS vs With OS"* comparison.

Answer:

An operating System is a software that helps the user interact with the computer and its hardware. Examples are Linux, windows, and macOS. Without an OS using a computer would be very difficult because the user would have toc control the hardware directly.

Problems a user face without an Operating System:

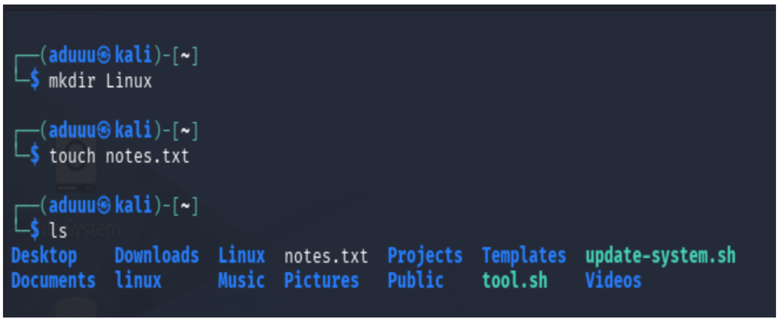
- Starting and using application would be very difficult.
- Users would have to communicate with the computer hardware directly.
- Creating and finding files would be harder.
- Managing RSM while several programs are running would be complicated.
- Controlling which users can access particular files or resources would be difficult.
- External hardware such as printers, USB devices and keyboards would be harder to use.

Without OS vs With OS:

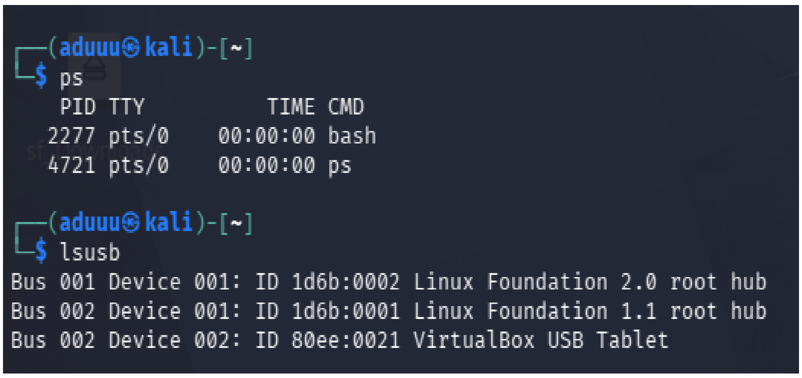
Without OS	With OS
Using computer would be difficult	Using computer would be easier
Managing files would be harder	We can create, move, delete files
Starting the program is difficult	We can open and use program easily
Handling the CPU would be harder	OS manages CPU for different tasks
Using printers and USB becomes difficult	OS helps to work with different devices

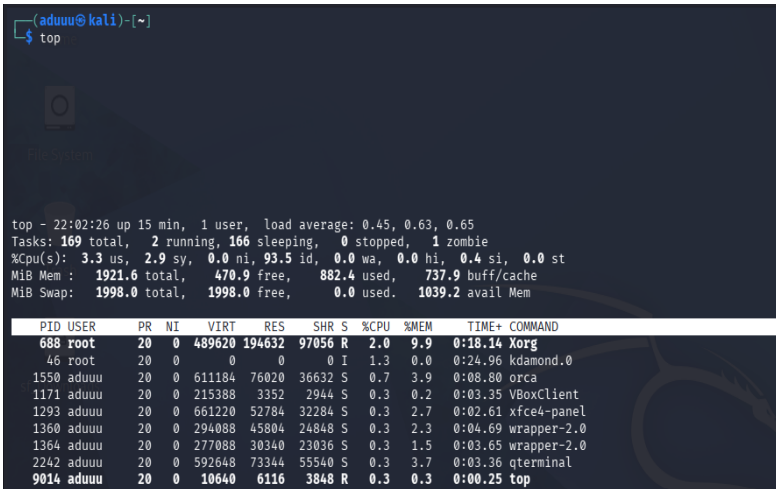
Linux Features That Solve These Problems

1. **File and Storage Management:**
   Linux provides commands for working with files and directories.

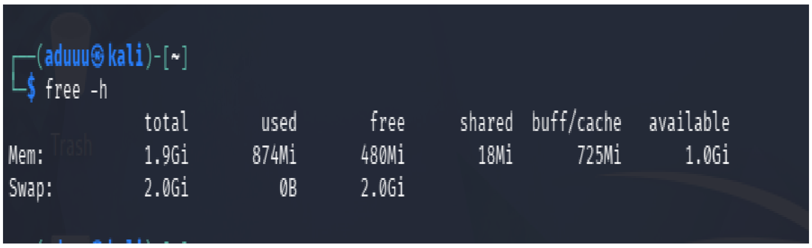


2. **Process Management:**
   Linux manages the programs running on a computer. It decides how much CPU and memory each program gets and lets users see or stop program when needed.

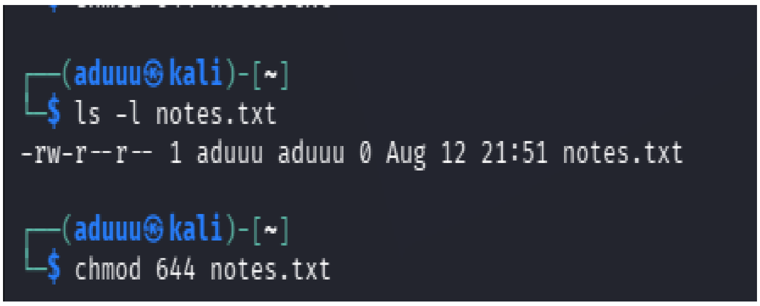




3. **Memory Management:**
   Linux manages the computer's memory and gives each program the memory it needs. If RAM is not enough then Linux will use space on the hard disk as extra memory.



4. **Security and User Management:**
   Linux protects multiple users and provide permission such as read write and execute.



2. Imagine your team is setting up a cybersecurity laboratory. Explore at least five Linux distributions relevant to cybersecurity and recommend suitable distributions for different roles such as penetration testing, digital forensics, privacy, and security monitoring. Present your findings creatively as a comparison chart, poster, infographic, or "OS selection guide."

Answer:

A cybersecurity laboratory uses different Linux systems for different security jobs. No single Linux distribution is best for every task, so we can choose ones for different purposes.

Linux Distribution	**Main use**	**Main Features**	**Recommended Role**
Kali Linux	Penetration Testing	Many hacking and security tools	Penetration Tester
Parrot Security OS	Testing and Privacy	Security, privacy, and forensic tools	Security Researcher and Privacy User
Tsurugi Linux	Digital Forensics	Forensics and malware investigation	Digital Forensics Investigator
Security Onion	Security Monitoring	Network monitoring and threat detection	SOC / Security Monitoring Analyst
Qubes OS	Privacy and Isolation	Strong privacy and safe isolation	Privacy and High-Security User

1.Kali Linux — Penetration Testing:

Kali Linux is a Linux distribution used for testing computer and network security. It provides many tools to find security weakness. It is mainly used by ethical hackers to test systems safely.

Best for: Penetration Testing

2. Parrot Security OS — Security & Privacy:

Parrot Security OS is used for cybersecurity, privacy and security testing. It provides tools for penetration testing and digital forensics. It is also suitable for secure everyday use.

Best for: Security and privacy

3. Tsurugi Linux — Digital Forensics:

Tsurugi Linux is designed for investigating digital evidence. It helps security professionals examine files, devices, and cyber incidents. It also includes a kernel level device write blocking feature which is useful when examining without accidently modifying it.

Best for: Digital Forensics

4. Security Onion — Security Monitoring:

Security Onion is used to monitor computer networks and detect security threats. It helps identify unwanted and suspicious activities and attack. It is mainly useful for security teams and SOC.

Best for: Security Monitoring

5.Qubes OS — Privacy & Isolation:

Qubes OS focuses on privacy and strong security. It keeps different activities separated to reduce security risk. This helps protect the system if one area is attacked. It also supports whonix integration for routing traffic through Tor.

Best for: Privacy and Isolation

3. Exit status:

Execute at least five Linux commands that produce both successful and unsuccessful results. Investigate their exit status using `$?`. Create a simple visual/table showing *Command → Result Exit Status → Meaning*. What can a script learn from an exit status?

Answer:

An exit status tells us that whether a Linux command worked successfully or not.

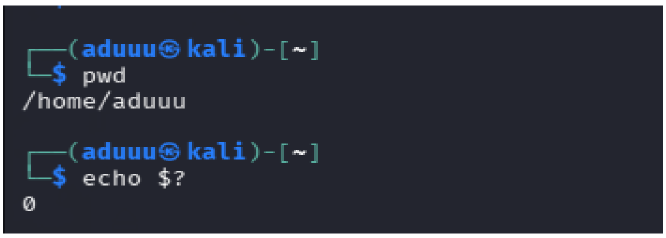
0 : command was successful.

Any non-zero number : command had an error.

Command → Result Exit Status → Meaning

Command	Result	Exit Status	Meaning
pwd	Shows current directory	0	successful
ls	Shows files and folders	0	successful
mkdir test	Create a folder	0	successful
ls abc	No such file or directory	2	error
Cat abc.txt	No such file or directory	1	error

1. **Command is successful:**



2. **Command is successful:**

```
┌──(aduuu㉿kali)-[~]
└─$ ls
Desktop  document  Documents  Downloads  linux  Linux  Music  notes.txt  Pictures  Projects  Public  Templates  tool.sh  update-system.sh  Videos

┌──(aduuu㉿kali)-[~]
└─$ echo $?
0
```

3. **Command is successful:**

```
┌──(aduuu㉿kali)-[~]
└─$ mkdir Practice

┌──(aduuu㉿kali)-[~]
└─$ echo $?
0

┌──(aduuu㉿kali)-[~]
└─$ ls Practice

┌──(aduuu㉿kali)-[~]
└─$ echo $?
0
```

4. **Command is unsuccessful:**

```
┌──(aduuu㉿kali)-[~]
└─$ ls abc
ls: cannot access 'abc': No such file or directory

┌──(aduuu㉿kali)-[~]
└─$ echo $?
2
```

5. **Command is unsuccessful:**

```
┌──(aduuu㉿kali)-[~]
└─$ cat abc.txt
cat: abc.txt: No such file or directory

┌──(aduuu㉿kali)-[~]
└─$ echo $?
1
```

What can a script learn from an exit status?

A script can use the exit status to know whether a command worked or not. An exit status of 0 usually means the command was successful. A non-zero status means that an error or problem occurred. The script can decide what to do next based on the result.